

Chaithanya Virupaksha

Machine Learning Engineer · PyTorch & Applied ML

Right to work UK Graduate Route from Dec 2026 · no sponsorship needed to start

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PROFILE

MSc Advanced Computer Science graduate (Newcastle University) who builds tested, reproducible machine-learning systems. My dissertation studied federated learning over lossy wireless networks: I built a PyTorch simulator modelling random and bursty packet loss and bandwidth limits, showed that gradient compression decides which devices can take part, and covered the framework with 50 unit tests. Before the MSc I spent 15 months full-time on client ML projects, from requirements through data preparation, training and evaluation to handover, working directly with the people who would use the results.

2.5 yrs industry ML, incl. 15 months full-time delivering client models	98% held-out accuracy, OpenCV object- detection pipeline	50 unit tests on a reproducible FL simulator	~25% Lighthouse performance gain from a front-end rebuild
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EXPERIENCE

Machine Learning Engineer May 2024 – Aug 2025

Accurate Info Solution · Bengaluru

- Built and delivered machine-learning models for client projects, mostly for engineering colleges, taking each from client requirements through data preparation, model training and evaluation to handover.
- Worked directly with clients to scope requirements, adapt models to their data and explain results to non-technical users.
- Progressed from intern to engineer, returning after graduation to own client ML projects end to end.

Full-Stack Development Intern Sep 2023 – Mar 2024

PACE Wisdom Solutions · Bengaluru

- Rebuilt front-end components in HTML, SCSS and Bootstrap on a component-based architecture, raising Lighthouse performance scores by around 25%.
- Shipped responsive features in two-week Agile sprints alongside design and back-end engineers.

SKILLS

CORE FOR THIS ROLE

- PythonPyTorchNumPy
- Pandasscikit-learn
- Unit testing
- Experimental design
- Model evaluation
- Federated learning
- Network simulationOpenCV
- CNNsNLP (TF-IDF, LDA)

LANGUAGES

- Java 21SQLJavaScript
- C++C

SOFTWARE

- FlaskRESTSpring Boot 3
- React 19PostgreSQLMySQL
- Microservices

CLOUD & DEVOPS

- Microsoft Azure
- Azure Kubernetes Service
- KubernetesDocker
- GitHub Actions CI/CD
- PrometheusGrafanaAWS S3
- Git

SECURITY

- OAuth2/JWTKeycloak
- Secure SDLCThreat modelling

Machine Learning Intern (part-time, alongside BEng)

Sep 2022 – Dec 2023

Accurate Info Solution · Bengaluru

- Built a Python and OpenCV object-detection pipeline across multiple object categories, reaching 98% classification accuracy on a held-out test set.
- Prototyped a CNN for plant-disease identification from leaf images that outperformed the rule-based baseline it replaced.

PROJECTS

Federated Learning under Lossy Wireless Networks

2026

MSc dissertation · Newcastle University · [GitHub repository](#)

- Built a federated-learning simulator, instrumented per device and per round, modelling random packet loss, bursty packet loss and bandwidth limits across gradient-compression schemes, four client-selection policies, MNIST, and IID and non-IID data.
- Found that compression does more than save bandwidth: it decides which devices can take part. The accuracy gap between selection strategies fell from 16.4 to 2.3 percentage points once updates arrived within the round deadline.
- Made every random draw a pure function of its inputs, so compared configurations saw identical network conditions; covered the framework with 50 unit tests.
- Audited my own work, found and fixed eight defects, and documented the ones that changed or qualified results in the repository changelog.

Python

PyTorch

Unit testing

NumPy

AI Bias Audit Framework for Generative AI

2025–2026

Newcastle University

- Built a framework that measures gender, cultural and role-stereotype bias across several LLMs using TF-IDF, LDA topic modelling and Shannon-entropy diversity scores, with a fixed prompt set so each run is a repeatable measurement.

Python

NLP

scikit-learn

Cloud-Native Bookstore Platform

2025–2026

Team of six · Newcastle University

- Built the React front end (catalogue, book detail, cart and checkout) for a microservices platform of Catalog, Cart and Gateway services on Azure Kubernetes Service.
- Co-authored the architecture report covering blue-green releases, autoscaling and Velero-based recovery.

Azure Kubernetes Service

React 19

Spring Boot 3

GitHub Actions

EDUCATION

MSc Advanced Computer Science

Newcastle University · 2025–2026

Dissertation submitted Sept 2026; award expected Dec 2026. Cloud Computing, Secure Software Development, Risk & Trust Management, Generative AI for Businesses, Enterprise Software & Services, Advanced Programming in Java.

BEng Computer Science

Dr. Ambedkar Institute of Technology, Bengaluru · 2024

CGPA 8.3 / 10

PUBLICATIONS

Virupaksha, C. (2026). Privacy-Preserving Digital Twin Models for Smart Healthcare Applications. *IJDTSC*, 2(1).

Siddaraju, Dr., Virupaksha, C. et al. (2024). Smart Irrigation Using Embedded AI. *IJIRCCCE*, 12(6).

CERTIFICATIONS

Python 3 Programming, University of Michigan
Databases and SQL for Data Science with Python

BEYOND CODE

Bharatanatyam performer with Newcastle University's Indian Performing Arts Society.

Organised Kalarava 2024, a college festival, coordinating volunteers and logistics across multiple events.

LANGUAGES

English (full professional) · Kannada (native) · Hindi (limited working)